

Steve Gwade

Full-Stack AI Engineer (AI-Driven Product & Systems)

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Professional Summary

Full-stack Engineer with 2+ years delivering production web and AI systems. Built end-to-end solutions across backend, frontend, and ML pipelines for clients in healthcare, travel, agriculture, and consumer sectors.

Experience

- Lead Backend Developer (Freelance)** Mar 2026 – May 2026
MITS SARM – B2B SaaS Travel Platform
 - Designed and built a conversational WhatsApp module for booking, as sole developer from architecture through deployment.
 - Implemented LLM-driven intent parsing, OTP email authentication, and bilingual support (FR/EN), reducing average booking request time by over 90% compared to the legacy email-based flow.
 - Built an approval pipeline with Redis-based concurrency control, eliminating duplicate approvals across concurrent sessions.
 - Integrated the module into an existing AWS event-driven backend without modifying any upstream services.
- Research Engineer (Full-Stack & Applied AI)** Jan 2025 – Jan 2026
JIANTS – Systems Intelligence Lab
 - Built AI-enabled internal platforms which led to the reduction of prototype-to-deployment time by approximately 30%.
 - Developed and deployed computer vision models for internal assistive applications.
 - Contributed to LINA, an assistive vision system for blind users, won 1st Prize worldwide at the Huawei ICT Competition.
 - Accelerated experimentation through AI-assisted coding and debugging workflows, improving iteration speed by over 70%.
- Software Engineer (Full-Stack & AI)** Mar 2024 – Dec 2024
TGNix – IT Consulting
 - Delivered client-facing web and AI solutions end-to-end from problem to product, improving operational efficiency and reducing manual processing time across multiple client usecases.
 - Built scalable backend APIs with Python, supporting stable performance under concurrent production workloads.
 - Integrated ML models into live systems, replacing manual decision processes with automated inference pipelines.
 - Deployed Dockerized applications to cloud platforms, reducing deployment errors and improving release reliability.
- Junior AI Backend Engineer (Intern)** Jun 2023 – Nov 2023
AITECAF – AI Technologies for Africa
 - Built REST APIs using Django REST Framework, enabling centralized access to AI and non-AI business services.
 - Developed chatbot backends using LangChain, reducing repetitive customer queries through automated responses.
 - Trained and deployed a YOLO-based cassava disease model enabling sub-5ms identification in agricultural workflows.
 - Strengthened model reliability by contributing to data annotation, evaluation, and testing pipelines prior to deployment.

Selected Projects

- VERO** (*Personal Project, Ongoing*)
 - Built a full-stack AI research workspace for ingesting documents, URLs, and public repositories into isolated project knowledge bases.
 - Designed a full-stack system with background parsing, indexing, persistent chat sessions, discovery views, and activity dashboards.
 - Implemented grounded answering with hybrid retrieval, reranking, and citations to keep responses traceable to source material.
- PlumVision** (*JCIA Hackathon Winner, 2025*)
 - Built a real-time plum classification system achieving 98% accuracy using a compact 3MB vision model.
 - Delivered sub-12ms inference latency by optimizing model architecture, preprocessing, and deployment pipelines.
 - Served real-time predictions through a FastAPI backend for live demos and application integration.
- EatWise** (*JCIA Most Innovative AI Prototype Winner, Ongoing*)
 - Developed a food recognition and nutrition analysis backend to support data-driven healthier eating decisions.
 - Integrated AI inference services into a mobile-oriented workflow connecting image input to nutrition insights.

Other Projects

- Spoof Detector** (*Personal Project*)
 - Trained a face spoofing detection model to strengthen biometric security in facial recognition pipelines.
 - Exposed the model via a Flask REST API for straightforward integration into authentication systems.
- Multi-Task Transfer Learning for Malaria Diagnosis** (*MSc Thesis, 2025*)

- Formulated a multi-task transfer learning objective, enabling joint optimization across heterogeneous tasks.
- Engineered data pipelines for task-specific datasets and align heterogeneous data distributions for multi-task training.
- Analyzed task interactions through experiments, highlighting trade-offs between single-task and multi-task training.
- Achieved a 97% macro F1 score, outperforming single-task baselines and demonstrating robust predictive performance.
- Deployed inference services using FastAPI to support reproducible evaluation and integration.

Technical Skills

- **Languages:** Python, JavaScript, SQL
- **Frontend:** React, React Native, HTML5, CSS3, Tailwind CSS, Bootstrap
- **Backend:** Django REST Framework, FastAPI, Flask, Node.js (Express)
- **Databases:** PostgreSQL, MongoDB, SQLite, Supabase
- **AI / ML:** PyTorch, TensorFlow, scikit-learn, LangChain, RAG pipelines
- **Cloud & Platforms:** AWS (ECS Fargate, SQS, DynamoDB, S3, SES), Google Cloud Platform, Firebase, Render
- **Data Stores:** Redis, ElastiCache
- **DevOps:** Docker, GitHub Actions, CI/CD pipelines, AWS (EC2, Fargate, S3, SageMaker)
- **Tools & Productivity:** Git, GitHub, Linux, Postman, Cursor, Google Stitch, Bolt

Awards and Recognition

- **2nd Place ODC Champions Africa 2025**, *Orange Digital Center* – 2nd among all African countries in an algorithmic coding contest on CodinGame with my team.
- **JCIA Hackathon 2025 Winner**, *PlumVision* – developed a real-time plum classifier achieving 98% accuracy, ~3MB model, <12ms inference, enabling instant quality control.
- **Most Innovative AI Prototype JCIA AI Solutions Track Winner**, *EatWise* – mobile AI assistant improving nutritional habits via food recognition, personalized meal plans, and hundreds of local recipes based on users' fridge, habits, and goals.
- **1st Prize Worldwide Huawei ICT Competition (Innovative Track)**, *LINA Assistive Vision System for the Blind* – contributed to building an AI system improving by 56% navigation, autonomy, and object recognition for blind users.

Academic Qualifications

- **MSc in Engineering** (2023–2025), National Higher Polytechnic School of Douala – specialization in Data Science & AI, MScEng thesis: multi-task transfer learning for malaria diagnosis
- **BSc in Engineering** (2020–2023), National Higher Polytechnic School of Douala – focus on Computer Science and foundations of Data Science & AI

Research & Publications

- **Multi-Task Transfer Learning and Application to Malaria Detection** – MSc Thesis, 2025, github.com/GwadeSteve/MTTL-Research-Malaria.

Professional Skills

- Problem solving, time management, critical thinking
- End-to-end product execution, ownership, stakeholder communication
- Translating business needs into technical architectures and APIs
- Efficient delivery using modern tooling and AI-assisted workflows

Interests

Reading, Gaming, Basketball, Coffee, Open-source contribution